

Whisper STT Lab Report

1. Differences between Prompted and Unprompted Transcriptions

The comparison shows that the guided (prompted) transcription performs slightly better than the basic (unprompted) version. Both versions capture the main content of the conversation, but the guided transcription produces clearer and more natural sentences.

The prompted version also handles Hawaiian terms such as *hukilau*, *lau*, and *akule* more consistently and integrates them better into the text. In contrast, the basic transcription contains more awkward phrasing and occasional misinterpretations.

Both approaches still make some errors. For example, both incorrectly transcribe “*lead*” as “*lid*”. However, the basic transcription introduces more incorrect or invented phrases, while the guided version has fewer and less impactful issues.

Overall, the guided transcription is more accurate and easier to read.

2. Benefits of Chunking for Long Audio

Chunking allows long audio files to be processed in smaller, manageable parts. This helps avoid API limits and reduces the risk of errors or timeouts when working with large recordings.

It also improves reliability, since each chunk is processed independently. If one chunk fails, it can be retried without reprocessing the entire file.

Another important benefit is better timestamp handling. By combining chunk results with adjusted offsets, the final transcript maintains a continuous timeline, making it easier to navigate and search.

3. Challenges Faced

The main challenge was prompt design. The initial prompts reduced transcription quality by including misleading or unnecessary vocabulary. This was resolved by refining the prompt to include only relevant and accurate terms based on the actual audio content.

Another challenge was aligning timestamps across chunks. Since each chunk starts at zero, timestamps needed to be adjusted manually. This was solved by adding the chunk’s start time to each segment, creating a consistent timeline across the full audio.

4. Recommendations for Improving Accuracy

To improve transcription accuracy, prompts should be concise and focused only on relevant vocabulary. Including too many assumptions or incorrect terms can reduce performance.

For specialized topics, it is helpful to include a small set of verified domain-specific words to guide the model.

Post-processing is also important. A second review step—either manual or using an LLM—can help correct errors, especially for technical terms.

Finally, combining chunking with guided prompts provides the best balance for long audio files, improving both reliability and overall transcription quality.